

REVIEW

What remains after Ozempic stops?

REFERENCES 15 verified	STYLE Popsai	MADE WITH Grounded v0.5.5	VERIFICATION Crossref · claims	COMPILED 5 Sep 2026
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Studies of semaglutide, the medicine in Ozempic, show that weight commonly returns and several metabolic improvements diminish after treatment ends. The extent of that reversal varies, and the evidence for preserving the benefits is less settled than the evidence for losing them.

Ozempic is an injectable brand of [semaglutide](#), introduced for type 2 diabetes.¹ A direct experiment on stopping the drug, however, studied a weekly 2.4 mg obesity regimen in adults without diabetes.² In the trial known as STEP 4, everyone began with semaglutide; after twenty weeks, 803 participants who had reached the study's maintenance dose were randomly assigned either to continue it or to switch to [placebo](#), an injection without the active drug.²

Over the next 48 weeks, mean weight changed by -7.9% with continued semaglutide and $+6.9\%$ after the placebo switch, both relative to week-20 weight.² Both groups continued lifestyle support, so the difference arose within a comparison that retained that part of treatment.² The common start and subsequent split in [Figure 1](#) express the central question: how much does keeping the drug change what happens next? Random assignment gives a strong answer for people who reached the maintenance dose, while that selection limits how fully the result describes everyone who starts semaglutide.²

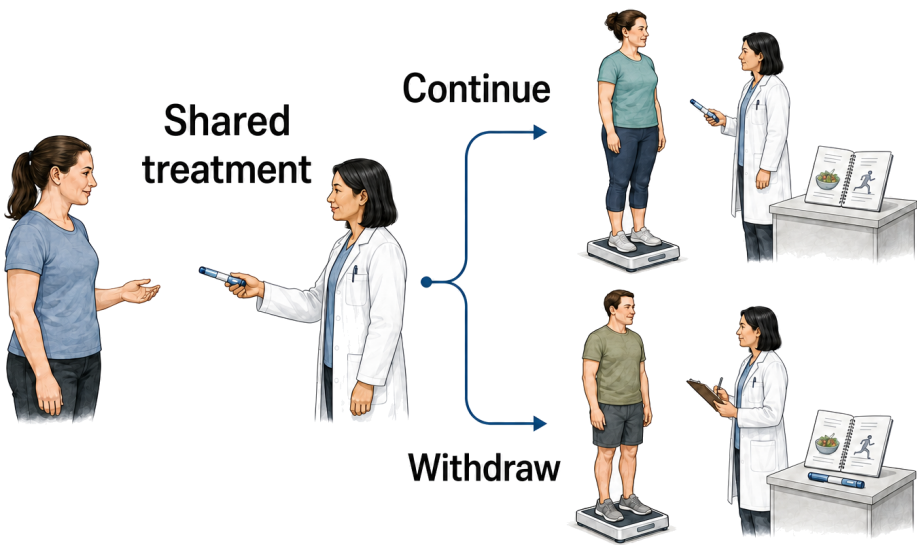


Figure 1. A shared start separates continuation from withdrawal. The branches depict STEP 4's random assignment after initial semaglutide treatment, with lifestyle support in both groups.² The later metabolic analysis used that randomized comparison.³ The drawn people and equipment do not encode individual outcomes.

01 Weight returns, but not uniformly

A longer treatment period was followed by substantial regain too. In the STEP 1 extension, participants stopped after 68 weeks of semaglutide or placebo and were followed for another year; unlike STEP 4, both the study medication and the lifestyle intervention ended.⁴ The extension analysis included 327 people from a selected subset of the original trial, so it provides a useful account of the following year without the same direct test of continuing versus stopping.⁴ Former semaglutide recipients had lost an average of 17.3% of their original weight.⁴ Over the year off treatment they

regained 11.6 percentage points of that original weight, roughly two-thirds of the preceding loss, and remained 5.6% below their starting weight on average.⁴ That residual loss also appeared in the individual measurements: 95 of the 197 former semaglutide recipients with weight data at the final visit still weighed at least 5% less than at the start.⁴ Substantial average regain therefore coexisted with retained benefit for many participants.

The two studies in [Figure 2](#) agree on the direction of change after withdrawal, but their vertical scales refer to different starting weights. STEP 4 expresses change relative

to weight at withdrawal; STEP 1 expresses regained weight as percentage points of the original baseline.^{2,4} Treatment length, participant selection and the end of lifestyle support

also differed, so the larger STEP 1 number cannot isolate the effect of stopping that support.^{2,4}

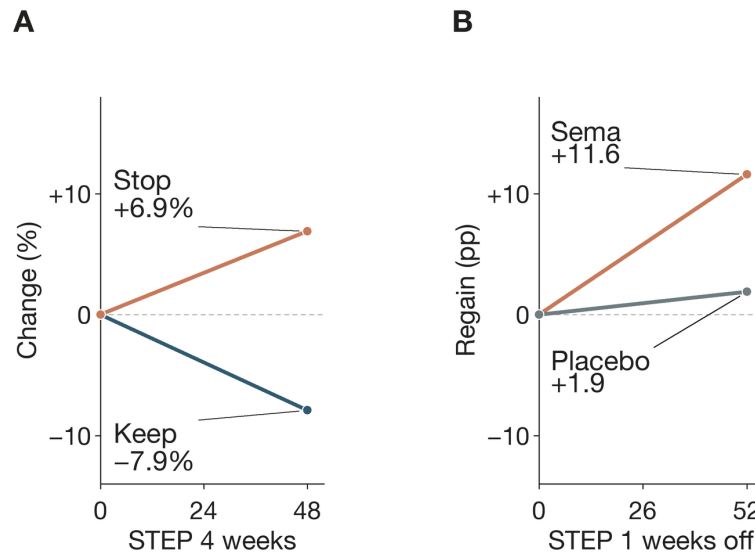


Figure 2. Both studies record regain after withdrawal. In panel A, the horizontal axis is weeks after STEP 4 randomization and the vertical axis is percent change from week-20 weight. “Keep” means continued semaglutide; “Stop” means the placebo switch. In panel B, the horizontal axis is weeks after STEP 1 treatment ended and the vertical axis is regain in percentage points (pp) of original baseline weight; zero marks withdrawal. “Sema” denotes former semaglutide recipients. Lifestyle support continued in STEP 4² and ended in STEP 1.⁴ Straight lines join endpoints rather than measured weekly paths; no uncertainty intervals are plotted.

02 The limits of a common rate

Broader reviews put these observations in context, while introducing another source of variation: different medicines and follow-up periods. A randomized-trial analysis including adults and adolescents found an average regain of 5.63 kg in participants with overweight or obesity, but differences between studies were extremely large.⁵ A separate

synthesis estimated 0.8 kg of monthly regain after semaglutide or tirzepatide.⁶ Its observations for those newer medicines extended no further than twelve months; the longer trajectory was projected.⁶ The table separates those evidence windows because an observed endpoint and a modelled rate answer different questions.

EVIDENCE	TIME AFTER STOPPING	WHAT THE ESTIMATE DESCRIBES
STEP 4	48 weeks	Randomized continuation versus withdrawal, with lifestyle support in both groups ²
STEP 1 extension	52 weeks	Follow-up after medication and lifestyle support ended; selected trial subset ⁴
Semaglutide/tirzepatide synthesis	Up to 12 months observed	Estimated regain of 0.8 kg/month (95% confidence interval 0.7–0.9); later course projected ⁶

Together, these findings establish common regain more convincingly than they establish a single rate. They also do not make the pooled rate a forecast for one person: the direct extension recorded varied outcomes, and even participants with larger initial losses tended to retain larger net losses despite regaining more.⁴

03 Appetite evidence comes from treatment

The weight measurements show what happened after stopping. Appetite studies help explain a possible reason, although they observe a different part of the process. Semaglutide leaves the body gradually: its long elimination half-

life supports weekly dosing, so exposure does not end with the final injection.¹ That establishes gradual clearance without fixing a date when hunger returns.

During treatment, people offered meals and snacks they could eat freely consumed about a quarter less energy with semaglutide than with placebo in one randomized experiment.⁷ A later trial using 2.4 mg also found less hunger, fewer or weaker cravings, and better control of eating.⁸ These findings identify appetite and intake effects that ongoing treatment provides. Losing these treatment effects is a plausible contributor to regain. The earlier experiment

did not map a day-by-day appetite trajectory after the last dose;⁷ neither did the later trial.⁸

The explanation also cannot be reduced to food remaining in the stomach longer. At week 20, the 2.4 mg trial detected no delay in stomach emptying using absorption of paracetamol as an indirect measure, despite the appetite effects.⁸ The test and timing leave room for earlier or individual stomach effects; they limit the claim that persistent slowing explains the observed appetite suppression.⁸ Evidence for eating less during treatment is firmer than evidence for exactly how eating changes after it ends.

04 Metabolic improvements also recede

Weight was accompanied by changes in other measurements. During STEP 4's shared treatment period, waist size, systolic blood pressure (the top reading), fasting blood sugar, insulin and blood fats improved.³ Those improvements were maintained with continued semaglutide but deteriorated after the switch to placebo.³ In the STEP 1 extension, most metabolic measures likewise moved back toward baseline, although some benefit remained after a year.⁴ These analyses support a common loss of benefit across several outcomes, with moderate confidence because many measurements were exploratory.

The diabetes evidence describes a different starting population. In a review pooling trials of semaglutide and related medicines, glycated haemoglobin, or HbA1c, rose after treatment stopped.⁵ The average increase was 0.25 percentage points in the obesity studies and 0.65 percentage points in studies of type 2 diabetes, with particularly wide variation between the diabetes studies.⁵ A selected diabetes cohort receiving continuing nutrition support retained improved blood sugar levels relative to its earlier baseline, although HbA1c rose during follow-up in both those who stopped and those who continued their medicine.⁹ The direction and amount of change depend partly on which population and treatment setting are being described.

Changes in these measures do not establish how long protection against heart attacks or strokes lasts. SELECT found fewer major cardiovascular events among high-risk adults without diabetes assigned to semaglutide than among those assigned to placebo.¹⁰ It was a treatment-assignment trial, not a randomized test of stopping after a common course.¹⁰ Its benefit and the withdrawal studies' worsening measurements therefore leave the duration of event protection after cessation unresolved.

05 Prescriptions follow less orderly paths

Outside trials, stopping is often followed by restarting. Among 125,474 US adults who began semaglutide or related medicines, about half of those with type 2 diabetes and almost two-thirds of those without it discontinued within a year; restarting was also frequent.¹¹ In a separate review of 288 patients' records, cost or insurance problems

were the most commonly documented reason for stopping semaglutide or tirzepatide, followed by side effects and shortages, while some records gave no reason.¹²

These interruptions help explain why routine-care studies ask a broader question than planned withdrawal trials. In a health-system cohort, people with available one-year weights who stopped within the first three months averaged 3.6% weight loss, compared with 11.9% among those without discontinuation.¹³ Both percentages describe change from starting treatment, rather than weight regained from the stopping date.¹³ The association is consistent with greater benefit during longer exposure, but dose, access and differences between people who continue and stop prevent attributing the entire gap to withdrawal.¹³

06 Maintenance differs across settings

Some settings show substantially more retained weight loss. Twenty-five women with polycystic ovary syndrome continued metformin and lifestyle support after a short course of semaglutide.¹⁴ Over two years, they regained about one-third of their semaglutide-associated loss, although several metabolic improvements returned to baseline.¹⁴ Their lower dose, brief treatment and absence of a control group distinguish this cohort from the obesity trials.¹⁴ It demonstrates a different observed trajectory without establishing that metformin caused the better maintenance.¹⁴

Similarly, a selected diabetes cohort maintained weight after medicines in semaglutide's drug class were stopped while carbohydrate-restricted nutrition care continued through telemedicine.⁹ The program's participants had not been randomly assigned to stop or continue.⁹ Such cohorts show that substantial regain is not universal, while leaving uncertain whether their results would transfer to other populations or follow from the accompanying intervention itself.

Exercise has been examined in a randomized treatment study, but with liraglutide, a related medicine, rather than semaglutide.¹⁵ After initial diet-induced weight loss, participants received exercise, liraglutide, both, or placebo for a year, followed by another year after treatment ended.¹⁵ The sequence in Figure 3 separates those periods because the study's main comparison covered both years together.¹⁵

Across that combined period, the exercise-plus-liraglutide group numerically retained more weight and body-fat loss than the liraglutide-only group.¹⁵ Those comparisons did not meet the study's prespecified threshold for accounting for multiple statistical tests.¹⁵ They also do not isolate better maintenance during withdrawal alone: for regain during the year off treatment, the uncertainty interval for combination versus liraglutide included no difference.¹⁵ Only 109 participants attended the later assessment, adding uncertainty from incomplete follow-up.¹⁵



Figure 3. Treatment and later maintenance occupy separate periods. The scenes show the combination arm's supervised exercise with liraglutide, the end of treatment, and assessment one year later.¹⁵ Arrows mark time, and body shapes do not represent measured weight changes. The primary endpoint spanned the treatment year and the following year together; it does not isolate an effect during withdrawal alone.¹⁵

Across a broader review of stopping weight-management medicines, there was no evidence that post-cessation behavioural support changed regain rates.⁶ That result does not establish that every programme is ineffective, but it prevents the selected cohort and liraglutide findings from becoming evidence of a broadly reliable semaglutide maintenance strategy. What persists after treatment is therefore partly clear: substantial weight regain and diminishing metabolic benefits are common, yet retained benefit varies. The evidence is strongest for the value of continued treatment in the studied population and less certain about which benefits can endure after stopping, for whom and for how long.

Scope and methods. This narrative review reuses the authenticated corpus searched on 4 September 2026 in

PubMed, with backward and forward OpenAlex citation chasing. No new literature search was conducted for this rewrite. It covers peer-reviewed human evidence on semaglutide cessation, focusing on adults while identifying mixed-age syntheses as broader context; related drugs and selected maintenance cohorts are identified explicitly, and anecdotes and preprints were excluded. Twelve full texts are authenticated in the saved access manifest; eleven are cited here. Remaining cited evidence was available at abstract level, including the direct STEP 4 report. Related trial reports and overlapping reviews are not counted as independent confirmation. Source support and confidence in the outcome remain separate judgments.

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58 CITED SENTENCES · 62 SOURCE CHECKS · 42 SUPPORTED AT FULL TEXT · 13 AT ABSTRACT · 7 PARTIAL · 0 CONTRADICTED

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